Lesson 5.9 - How to Calculate the Expected Value

> **Video:** How To Calculate Expected Value by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

Formula

The **expected value** is the long-run **average** result per trial:

$$E(X) = \sum x\,P(x) = x_1p_1 + x_2p_2 + \cdots + x_np_n$$

- x = the value of each outcome (gains positive, losses **negative**)
- p = its probability, as a **decimal**; all the p's add up to 1

Steps: make a table of *outcome* → *value* → *probability*, multiply each value by its probability, and add.

Expected total over n trials = $n \times E(X)$.

Example 1 - a casino game

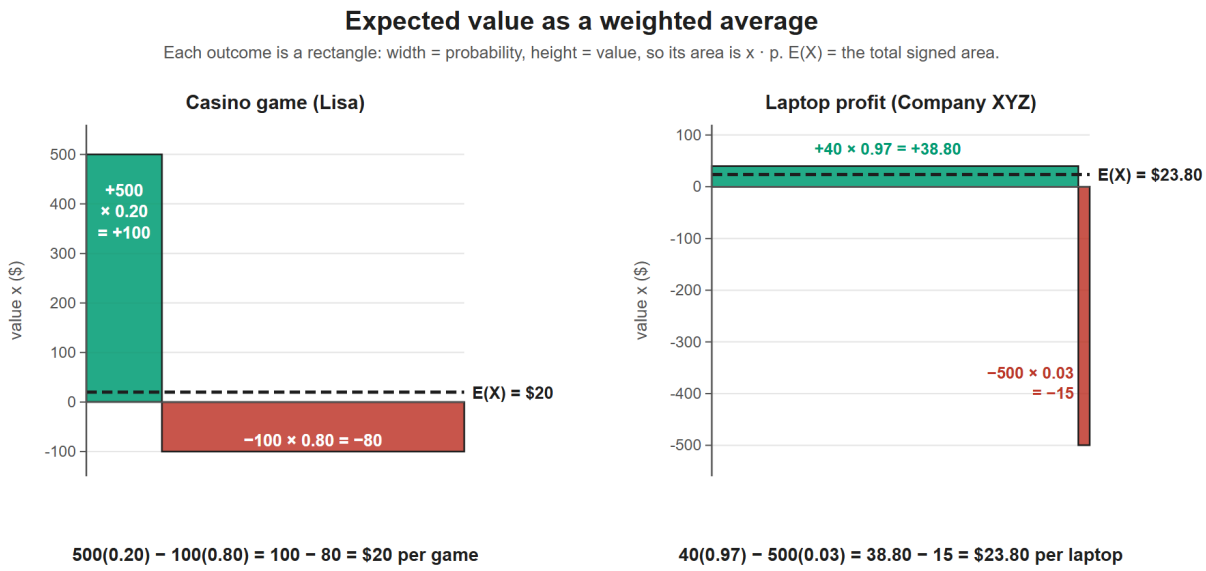
Lisa wins **$500** with probability 20%, and otherwise loses **$100**.

Outcome	Value x	Probability p	$x \cdot p$
Win	+500	0.20	+100
Lose	−100	0.80	−80
Total		**1.00**	**+20**

$$E(X) = 500(0.20) + (-100)(0.80) = 100 - 80 = \$20 \text{ per game}$$

- Over 10 games: $10 \times 20 = \$200$ expected.
- Over 100 games: $100 \times 20 = \$2{,}000$ expected.

(The probability of losing comes from the complement: $1 - 0.20 = 0.80$.)



Each outcome becomes a rectangle with area x times p. Positive areas (green) minus negative areas (red) give E(X). The dashed line at E(X) is the average height over the whole width.

Example 2 - profit per laptop

Company XYZ earns **$40** on each working laptop it sells and loses **$500** on each one returned as defective. **3 in 100** laptops are defective.

Outcome	Value x	Probability p	$x \cdot p$
Works	+40	0.97	+38.80
Defective	−500	0.03	−15.00
Total		**1.00**	**+23.80**

$$E(X) = 40(0.97) + (-500)(0.03) = 38.80 - 15.00 = \$23.80 \text{ per laptop}$$

Selling 100 laptops is expected to earn $100 \times 23.80 = \$2{,}380$.

Key takeaways

- $E(X) = \sum x\,p$: weight each outcome's value by its probability.
- Losses go in as **negative** values.
- The expected value is an **average over many trials**. A single game never pays exactly $20.
- Total expected = number of trials × expected value per trial.